

OpTC chronological online detection, reconstruction, and cost

3,002 online checkpoints | 15,010 frozen model responses | 76 observable official attack steps | 73 official adjacent transitions | 207 auxiliary held-out windows

Method	Aux. balanced accuracy	Attack windows correct	Background windows correct	Official steps recovered	Coverage-adjusted timeline	Official transition F1	Cost / 3,002 checkpoints	Avg. models / checkpoint	Saving vs. online full
DeepSeek V4 Flash	52.5%	6/35	151/172	48/76 (63.2%)	33.2%	7.2%	\$2.76	1.00	96.3%
DeepSeek V4 Pro	52.2%	15/35	106/172	51/76 (67.1%)	34.2%	9.8%	\$9.09	1.00	87.7%
Gemini 3.5 Flash-Lite	58.6%	30/35	54/172	49/76 (64.5%)	32.3%	5.2%	\$10.67	1.00	85.6%
GLM 5.2	60.9%	30/35	62/172	49/76 (64.5%)	34.1%	11.8%	\$31.13	1.00	58.0%
MiniMax M3	58.9%	29/35	60/172	49/76 (64.5%)	35.0%	11.8%	\$20.38	1.00	72.5%
Online-weighted full vote	65.9%	15/35	153/172	49/76 (64.5%)	35.0%	14.0%	\$74.03	5.00	0.0%
CaMVo (delta=0.80, kmin=3)	64.8%	12/35	164/172	49/76 (64.5%)	35.0%	14.0%	\$22.66	3.01	69.4%

Reading the table. The professor-aligned reconstruction endpoints use 76 official observable red-team steps and 73 adjacent official transitions. The 207-window class-balanced score is an auxiliary detection endpoint; all 35 attack and 172 background windows are retained, with each class contributing 50%.

Main result. CaMVo matches the online-weighted complete panel on official-step recall (49/76), coverage-adjusted timeline (35.0%), and transition F1 (14.0%), while reducing replay-policy cost from \$74.03 to \$22.66 (69.4%). Its auxiliary balanced accuracy is 64.8% versus 65.9% for the full panel and 60.9% for the strongest single model. DeepSeek V4 Pro recovers two more isolated official steps (51/76), but has lower timeline coverage, transition F1, and auxiliary balanced accuracy.

Uncertainty and scope. Step-recall Wilson intervals and campaign-aware paired bootstrap results are included in the numerical/reproducibility records. OpTC contains only three top-level campaigns, so campaign-level uncertainty is reported as sensitivity and no claim of 76 independent campaigns is made. Costs use recorded tokens and the frozen price snapshot and count only calls each policy would request.